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DISTRIBUTED SYSTEMS PROJECT 1

CUSTOM RELIABILITY PROTOCOL

Four-Way Handshake

- Client sends data with a unique Request ID (RID)
- Server acknowledges receipt and assigns/confirms a Client ID
- Server sends the computed result
- Client confirms receipt of the result so the server can clear its cache

FAULT TOLERANCE & CRASH RECOVERY

- **Persistent Client Store:** The server saves client states to a binary file. If the server reboots, it reloads the store and notifies "Processing" clients to reset.
- **Client State Persistence:** The Python library maintains a ``client_state.json``. If the client crashes mid-request, it recovers upon restart and sends a ``CANCEL_SVC`` to the server to clean up "zombie" tasks.
- **Heartbeat Monitoring:** During long-running tasks (like prime number calculations), the server pings the client to ensure it is still "alive."

SYSTEM

ARCHITECTURE

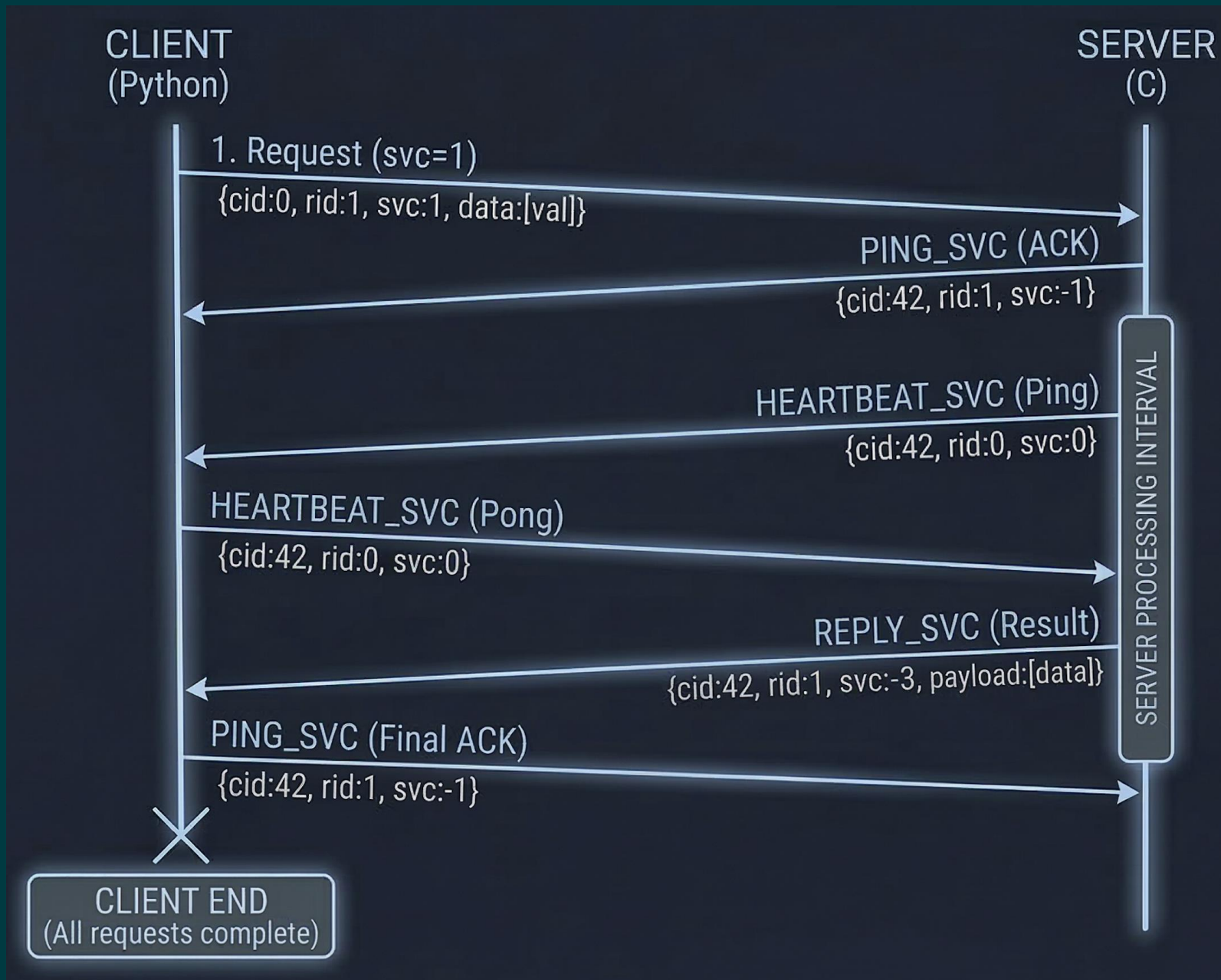
SERVER (C)

Four specialized background threads

- **Receiver:** Listens for all incoming packets; manages the Client Store and routes requests to per-service queues
- **Sender:** Waits on the reply queue and delivers results back to clients
- **Maintenance:** Runs every 2 seconds to send heartbeats to `PROCESSING` clients and retransmit replies to `WAITING\_ACK` clients. Marks clients `EXPIRED` after `MAX\_RETRIES` timeouts
- **Discovery:** Listens on port `9999` and responds to broadcast pings so clients can auto-discover the server

CLIENT (PY)

- Main Thread:
 - Initiates requests
 - Handles the high-level logic (Load Balancing, Handshaking)
 - Blocks until the result is ready
- Dispatcher Thread:
 - Constantly listens to the socket.
 - Puts incoming packets into a queue for the Main Thread to process.
 - Prevents Packet Loss: Ensures the OS buffer doesn't overflow while the main thread is busy.



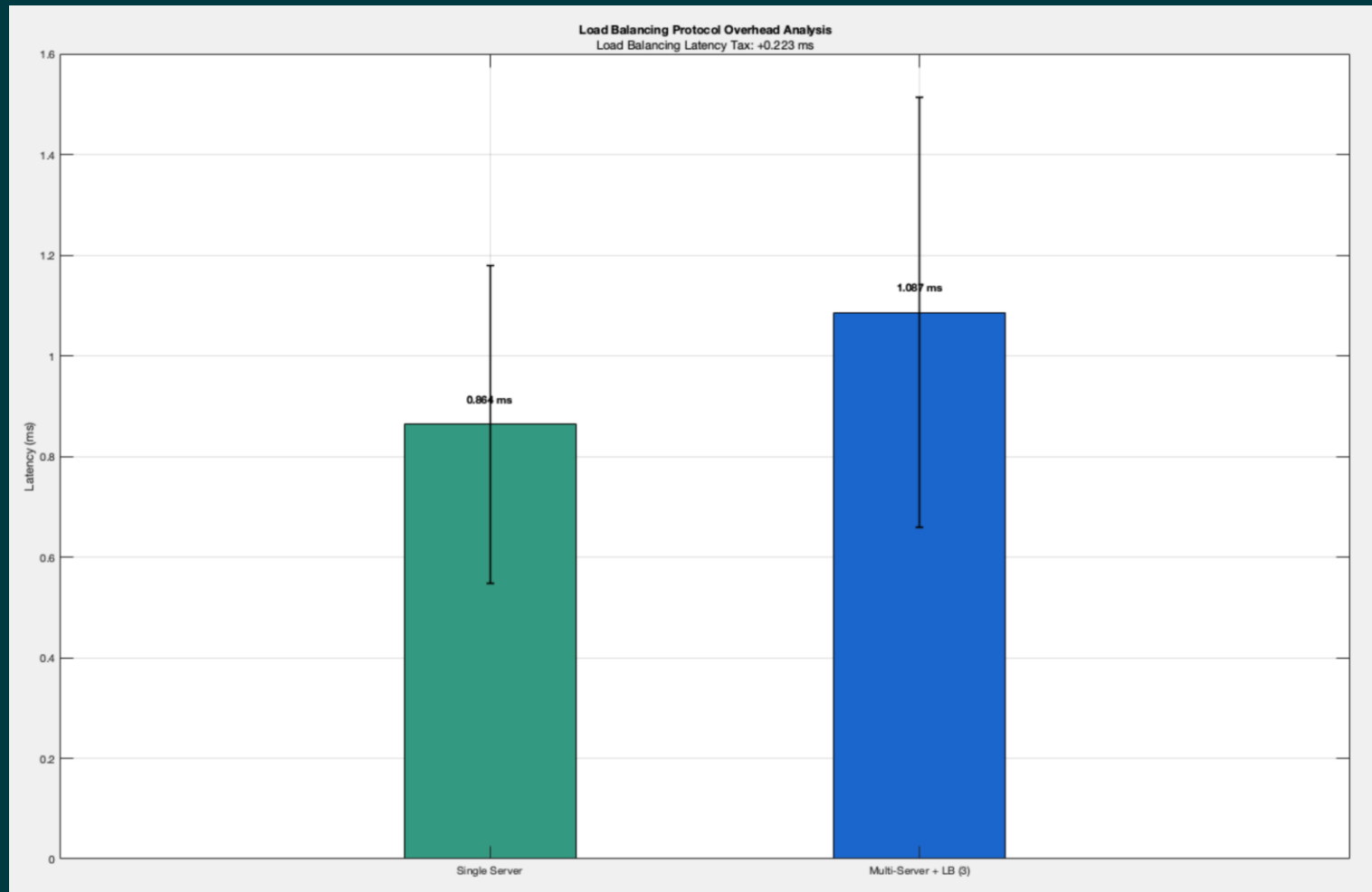
CONTROL SERVICE IDS

- **PING_SVC:** Server → Client: heartbeat during "PROCESSING". Client → Server: pong response
- **ERROR_SVC:** Server signals an error (invalid service, full table, stale request, etc.)
- **REPLY_ACK_SVC:** Server → Client: carries the final result. Client → Server: final ACK to confirm receipt.
- **CANCEL_SVC:** Client cancels an in-flight request (used on crash recovery)
- **LOAD_BALANCE_SVC:** Client → Server: asks for load info (service ID in reqid). Server → Client: replies with active_count in the cid field and pending_count in the reqid field (len = 0)
- **DISCOVERY_SVC:** Client broadcasts on port 9999; server echoes back so the client can discover its address.

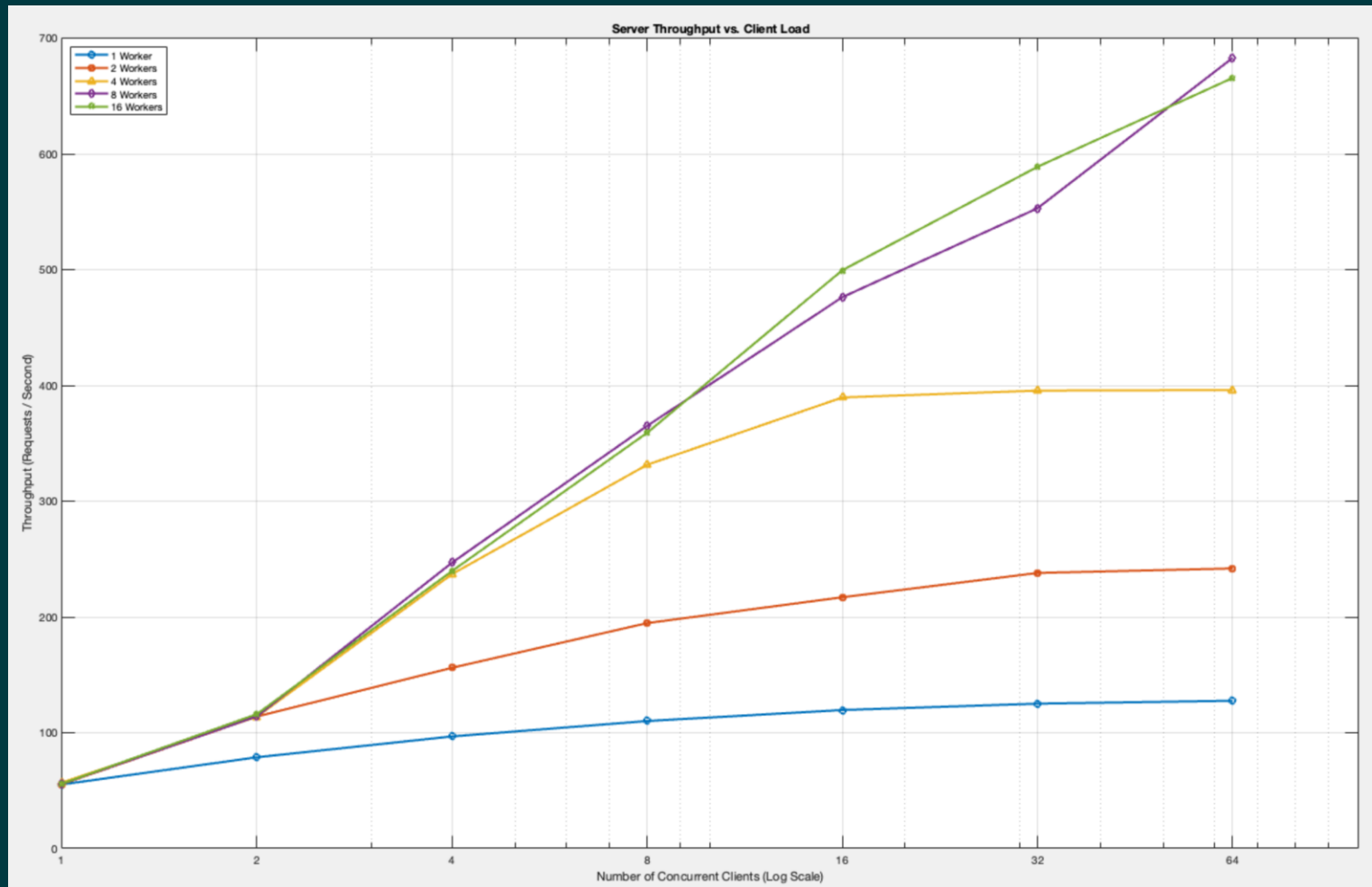
LOAD BALANCING

- The client queries all known servers for their load and routes work to the optimal one. Each server reports two counters:
 - Pending: requests queued, not yet picked up by a worker
 - Active: requests currently being processed
- The client selects the server with the lowest pending count, using active count as a tiebreaker.

NULL REQUEST-REPLY



SERVER REQUEST THROUGHPUT



SERVER AVG PROCESSING TIME

